Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

Continuous Process Improvement with FMEA

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TABLE OF CONTENTS

How to Use This Guide **4**
Connected Case Study **4**
Learning Outcomes and Competency Mapping **4**
Knowledge 4
Abilities 4
Topic 1 — Introduction to Continuous Process Improvement **5**
Topic Orientation 5
Concept Guide 5
Activity 01 — Launch the Orion Continuous Improvement System 7
Activity 02 — Define the Process with SIPOC and PDCA 9
Topic 2 — Root Cause Analysis **11**
Topic Orientation 11
Concept Guide 11
Activity 03 — Prioritise the Defect Pattern with Pareto Analysis 13
Activity 04 — Verify Root Causes with Fishbone, 5 Whys and Fault Tree 15
Topic 3 — Design and Apply FMEA **17**
Topic Orientation 17
Concept Guide 17
Activity 05 — Build the Orion Process FMEA 21
Activity 06 — Rate Risk, Plan Actions and Re-score 23
Topic 4 — Identify and Direct Improvement Opportunities **25**
Topic Orientation 25
Concept Guide 25
Activity 07 — Select the Improvement Portfolio 28
Activity 08 — Pilot, Direct and Coach the Change 30
Topic 5 — Manage Improvement with a Process Control Plan **32**
Topic Orientation 32
Concept Guide 32
Activity 09 — Evaluate Stability and Capability 35
Activity 10 — Create the Process Control Plan and Review Loop 37
Assessment Readiness **39**
References **39**

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How to Use This Guide

Use the concept chapters before each activity. Complete the Starter workbook in the matching activity folder, record evidence and assumptions, then use the checklist. The Completed workbook is for comparison after your own attempt.

Connected Case Study

Orion Precision Electronics (OPE). Surface-mount technology (SMT) assembly of industrial sensor-controller boards.

Rising solder defects, rework, late lots and two customer returns after a new product ramp-up.

The goal is to reduce first-pass defects, stabilise cycle time and prevent recurrence through a living PFMEA and Control Plan. Every activity advances the same case, so later decisions must trace to earlier evidence.

Learning Outcomes and Competency Mapping

- ELO1: Create a continuous process improvement system.
- ELO2: Perform root cause analysis for continuous process improvement.
- ELO3: Design and apply Failure Mode and Effects Analysis (FMEA).
- ELO4: Identify and direct improvement opportunities.
- ELO5: Manage and evaluate improvement with a Process Control Plan.

Knowledge

- K1: Identification of continuous improvement opportunities
- K2: Direct continuous improvement activities
- K3: Management of systems and processes to facilitate continuous improvement
- K4: Monitoring and evaluation of continuous improvement processes

Abilities

- A1: Manage and implement continuous improvement systems
- A2: Promote team support and coach team members toward continuous improvement
- A3: Gather, access and record production information to track continuous improvements
- A4: Analyse, synthesise and interpret information
- A5: Design and apply improvement tools and strategies
- A6: Monitor and evaluate improvement processes
- A7: Identify continuous improvement opportunities
- A8: Discuss and direct continuous improvement activities
- A9: Manage systems and processes to facilitate continued improvement
- A10: Evaluate improvement processes

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Topic 1 — Introduction to Continuous Process Improvement

Alignment: ELO1 · A1 · A2 · K1

Focus: Culture · governance · PDCA · process definition

Topic Orientation

System, not event

Continuous improvement is a managed cycle of evidence, action, learning and standardisation.

People closest to work

Operators and technicians surface weak signals and practical constraints.

PDCA discipline

Test a defined change, compare evidence and standardise only what works.

Shared process view

SIPOC and a process map establish boundaries before analysis begins.

Concept Guide

Improvement is a management system

A workshop creates ideas; a system creates repeatable learning, ownership and control.

What continuous improvement changes

Performance: Reduce defects, delay, cost and instability against a defined baseline.

Capability: Build reliable processes that repeatedly meet customer and regulatory needs.

Learning: Turn experiments, failures and operator knowledge into standard work.

Culture: Make problem visibility safe and improvement part of normal work.

Continuous improvement and breakthrough change

First perspective: Frequent, evidence-led adjustments; Lower implementation risk; Builds habits and process knowledge.

Contrast or design check: Larger redesign or technology shift; Higher uncertainty and coordination; May be necessary when the current system is structurally incapable.

Lean, Six Sigma and FMEA

Lean: Improve flow; Remove non-value work; Shorten lead time.

Six Sigma: Reduce variation; Use data to test causes; Improve capability.

FMEA: Anticipate failure; Prioritise prevention; Link risk to controls.

A culture that can improve

Will: Leaders make priorities and trade-offs explicit.

Skill: Teams know how to frame, analyse, test and control.

Authority: Owners can act within agreed boundaries.

Guidance: Coaching and standards turn tools into good judgement.

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Continuous improvement governance

Sponsor sets direction → Process owner defines gap → Cross-functional team analyses → Action owners implement → Governance reviews evidence

PDCA turns change into learning

Plan: define gap, target and hypothesis → Do: test at controlled scale → Check: compare evidence and side effects → Act: standardise, adjust or abandon

Plan with evidence

Problem: Current versus required performance, with scope and time.

Baseline: Trusted operational data and agreed definitions.

Target: Specific outcome, measure and time horizon.

Hypothesis: Why the proposed countermeasure should change the result.

Team roles in improvement

Sponsor: Removes barriers and protects the business priority.

Process owner: Owns performance and the post-project control system.

Facilitator: Keeps the method rigorous and participation balanced.

Operators and experts: Provide actual-work knowledge and test feasibility.

Data owner: Protects definitions, lineage and evidence quality.

Action owner: Implements, verifies and documents a countermeasure.

SIPOC establishes the analysis boundary

Suppliers → Inputs → Process → Outputs → Customers

Voice of customer and voice of process

First perspective: Needs, complaints and field effects; Critical-to-quality expectations; What failure means to the user.

Contrast or design check: Yield, cycle time and defect mechanisms; Variation, controls and capability; What the operation can repeatedly deliver.

Coaching for team support

Ask before telling: Use questions to surface assumptions and evidence.

Make work visible: Use boards, owners and dates so risk is shared.

Reinforce learning: Review what changed and why, not only the result.

Protect candour: Treat an exposed weakness as useful information.

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Activity 01 — Launch the Orion Continuous Improvement System

Mapping	A1, A2, K1
Objective	Create governance, coaching and visible management for the improvement effort.
Deliverable	CI charter, role map, coaching cadence and daily/weekly review board
Tools/files	Excel charter workbook · CI coaching checklist · PDCA governance

Scenario

Orion's SMT line has competing priorities, informal escalation and no shared definition of success.

Detailed Step-by-Step Procedure

Step 1 — Read the Orion case brief and list the customer, business and process consequences of the current defects.

Work in: Case Brief.pdf

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Define the problem without naming an unverified cause: process, period, measure, baseline and impact.

Work in: CI-System.xlsx · Charter

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Set the 90-day goal for first-pass yield, defect rate and cycle time; record units and owners.

Work in: CI-System.xlsx · Measures

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Assign sponsor, process owner, facilitator, operators, quality, maintenance and data-owner responsibilities.

Work in: CI-System.xlsx · Roles

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Create daily, weekly and monthly review cadences with escalation rules and evidence required.

Work in: CI-System.xlsx · Governance

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — Use the coaching checklist to plan how the facilitator will build team support and psychological safety.

Work in: Coaching-Checklist.md

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Scenario Questions

1. Which decision belongs to the sponsor rather than the team?
2. How will the process owner know the improvement is sustained?

Acceptance Criterion

A reviewer can identify the gap, target, measures, owners, cadence and coaching approach without asking for missing definitions.

Submission Check

- □ All required tabs or sections are complete.
- □ Formula results reconcile to their source data.
- □ Evidence and assumptions are traceable.
- □ Both scenario questions are answered.
- □ The workbook filename clearly identifies the learner.

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Activity 02 — Define the Process with SIPOC and PDCA

Mapping	A1, A2, K1
Objective	Set a process boundary and convert the improvement goal into a testable PDCA plan.
Deliverable	SIPOC, five-stage SMT process map, stakeholder needs and PDCA experiment canvas
Tools/files	SIPOC web tool · Excel workbook · PDCA hypothesis

Scenario

The team disagrees about whether stencil preparation, printing, placement or reflow belongs in scope.

Detailed Step-by-Step Procedure

Step 1 — State the start and end boundary: released kit and approved program through inspected board ready for test.

Work in: SIPOC-PDCA.xlsx · Scope

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Identify suppliers and inputs, including paste, stencil, boards, programs, maintenance and work instructions.

Work in: SIPOC-PDCA.xlsx · SIPOC

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Map the five macro steps and define one output/requirement for each.

Work in: SIPOC-PDCA.xlsx · Process Map

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Capture customer and next-operation requirements for first-pass quality, cycle time and traceability.

Work in: SIPOC-PDCA.xlsx · Requirements

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Write a PDCA hypothesis linking one proposed process change to a measurable result and guardrail.

Work in: SIPOC-PDCA.xlsx · PDCA

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — Review scope with an operator and record one assumption that needs evidence.

Work in: SIPOC-PDCA.xlsx · Review

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Scenario Questions

3. What is deliberately out of scope?
4. Which requirement will later appear as a function in the PFMEA?

Acceptance Criterion

The SIPOC boundary is unambiguous and the PDCA hypothesis contains change, predicted outcome, measure and guardrail.

Submission Check

- □ All required tabs or sections are complete.
- □ Formula results reconcile to their source data.
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Optional tool: SIPOC Builder

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Topic 2 — Root Cause Analysis

Alignment: ELO2 · A3 · A4

Focus: Evidence · Pareto · Fishbone · 5 Whys · Fault Tree

Topic Orientation

Describe the gap

A good problem statement names the measure, location, timing and magnitude without guessing the cause.

Separate cause from symptom

The customer effect, process failure and causal mechanism belong at different levels.

Triangulate

Use data prioritisation, structured brainstorming and verification together.

Mechanism over blame

Useful causes describe how the process creates failure and point toward controllable action.

Concept Guide

A root cause is a verified mechanism

It is not the first plausible explanation, a correlation, or the person nearest the failure.

Build a neutral problem statement

What: The measurable defect, delay or performance gap.

Where: Process step, line, product family or customer segment.

When: Start, pattern, shift, period or change point.

How much: Magnitude, rate, trend and business/customer impact.

Cause, failure and consequence

First perspective: Cause: stencil aperture partly blocked; Mechanism: insufficient paste transfer; Controllable through cleaning and verification.

Contrast or design check: Failure mode: insufficient solder joint; Effect: intermittent signal or field return; Observed symptom is not automatically the root cause.

Pareto focuses the investigation

Illustrative defect mix: Solder bridge 42, Insufficient solder 26, Misalignment 15, Wrong component 9, Tombstone 5, Other 3. The categories sum to 100; use the cumulative percentage to focus investigation without claiming the largest category is a root cause.

Read a Pareto chart correctly

Rank: Bars order categories by frequency or impact.

Cumulative line: Shows how quickly the categories account for the total.

Stratify: Check shift, product and machine before combining data.

Do not overclaim: The largest category is a priority, not yet a cause.

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Fishbone structures possible causes

People → Machine → Method → Material → Measurement → Environment

From brainstorm to evidence

Candidate cause: Write a specific mechanism, not a vague label.

Prediction: State what data pattern would be expected if it were true.

Test: Compare lots, shifts, machines, materials or controlled trials.

Decision: Confirm, reject or keep open with the next evidence needed.

Five Whys follows one causal chain

Observed failure → Immediate mechanism → Process condition → System weakness → Controllable root cause

Strong and weak 5 Whys

First perspective: Each answer is evidence-supported; The chain stays on one mechanism; The endpoint suggests a system change.

Contrast or design check: Jumps between unrelated branches; Ends with 'operator error'; Treats five as a magic number.

Fault Tree Analysis tests combinations

Top event → OR: any branch can cause it → AND: conditions must combine → Basic events with evidence

Fishbone, 5 Whys and Fault Tree

First perspective: Fishbone widens the search; 5 Whys deepens one likely chain; Use when mechanisms are still being explored.

Contrast or design check: Fault Tree formalises logical combinations; Supports safety and reliability reasoning; Use when several paths can create the top event.

Data quality checks before conclusion

Definition: Are defect categories and denominators consistent?

Measurement: Can inspectors or sensors reproduce the classification?

Completeness: Are rework, scrap and escaped defects all represented?

Context: Are product mix, shift and process changes recorded?

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Activity 03 — Prioritise the Defect Pattern with Pareto Analysis

Mapping	A3, A4
Objective	Gather, validate and interpret production information to focus the investigation.
Deliverable	Clean defect log, frequency summary, cumulative percentage and interactive Pareto chart
Tools/files	Excel Table · COUNTIFS · SUM · cumulative percentage · Pareto chart

Scenario

Orion has 240 inspection records across products, shifts and printing machines but anecdotes point in different directions.

Detailed Step-by-Step Procedure

Step 1 — Open the Data Dictionary and confirm one record per inspected lot, field definitions and allowed defect categories.

Work in: Defect-Pareto.xlsx · Data Dictionary

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Check blanks, duplicate Lot IDs, inconsistent defect labels and impossible counts; document each correction.

Work in: Defect-Pareto.xlsx · Data Quality

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Summarise defect counts by category with formulas and calculate each category's percentage of the total.

Work in: Defect-Pareto.xlsx · Pareto

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Calculate cumulative percentage and verify the final row equals 100%.

Work in: Defect-Pareto.xlsx · Pareto

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Create a descending column chart with a cumulative line on the secondary axis.

Work in: Defect-Pareto.xlsx · Pareto Chart

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — Stratify the leading defect by machine, shift and product family; write a bounded investigation statement.

Work in: Defect-Pareto.xlsx · Stratification

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Scenario Questions

5. What is the leading defect and how much of the total does it represent?
6. Which stratum shows the strongest signal, and what limitation remains?

Acceptance Criterion

The Pareto counts reconcile to the source, cumulative percentage reaches 100%, and the conclusion distinguishes priority from cause.

Submission Check

- □ All required tabs or sections are complete.
- □ Formula results reconcile to their source data.
- □ Evidence and assumptions are traceable.
- □ Both scenario questions are answered.
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Optional tool: Pareto Chart Builder

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Activity 04 — Verify Root Causes with Fishbone, 5 Whys and Fault Tree

Mapping	A3, A4
Objective	Analyse evidence and build a defensible causal explanation for the leading defect.
Deliverable	6M Fishbone, evidence-backed 5 Whys, fault tree and cause-verification matrix
Tools/files	Fishbone tool · 5 Whys tool · Fault Tree · verification planning

Scenario

Solder bridging is concentrated on Printer P2 during the late shift, but the causal mechanism is not yet verified.

Detailed Step-by-Step Procedure

Step 1 — Write the effect at the fish head: excess solder forms a conductive bridge after reflow.

Work in: RCA-Workbook.xlsx · Fishbone

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Brainstorm specific mechanisms under People, Machine, Method, Material, Measurement and Environment.

Work in: RCA-Workbook.xlsx · Fishbone

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Select the most evidence-supported branch and build a 5 Whys chain without changing branches midstream.

Work in: RCA-Workbook.xlsx · 5 Whys

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Build a fault tree for the top event and use AND/OR logic to show alternative and combined pathways.

Work in: RCA-Workbook.xlsx · Fault Tree

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — For each candidate cause, state the expected data signature, evidence source, test and decision rule.

Work in: RCA-Workbook.xlsx · Verification

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — Record confirmed, rejected and open causes; propose corrections only for confirmed or high-risk open causes.

Work in: RCA-Workbook.xlsx · Findings

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Scenario Questions

7. Which cause is a mechanism rather than a label?
8. What observation would disconfirm the preferred explanation?

Acceptance Criterion

At least one cause is supported by a reproducible evidence test, the fault-tree logic is valid, and no conclusion relies on blame.

Submission Check

- □ All required tabs or sections are complete.
- □ Formula results reconcile to their source data.
- □ Evidence and assumptions are traceable.
- □ Both scenario questions are answered.
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Optional tool: Fishbone Builder

Optional companion tool: 5 Whys Builder

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Topic 3 — Design and Apply FMEA

Alignment: ELO3 · A5 · A6 · K3

Focus: Scope · function · failure chain · risk · action · learning

Topic Orientation

Prospective thinking

FMEA asks what could fail before the next customer or process loss occurs.

Failure chain

Function → failure mode → effect → cause → controls makes risk traceable.

Risk-informed action

Severity, occurrence and detection guide judgement; RPN alone never grants permission to ignore risk.

Living knowledge

Actions, verification, re-rating and review triggers keep the FMEA current.

Concept Guide

FMEA is prevention made visible

It documents what might fail, why it matters, what controls exist and who will reduce the risk.

Source note: ASQ defines FMEA as a systematic, proactive method for evaluating where and how a process might fail and the relative impact of those failures.

When FMEA has the most leverage

New design or process: Prevent known patterns before approval or launch.

Changed use: Reassess risk when a product or process enters a new context.

Improvement goal: Anticipate failure introduced by the proposed change.

Before Control Plan: Convert prioritised risks into prevention and detection controls.

Three common FMEA views

System FMEA: Interfaces and interactions; System functions; Cross-boundary effects.

Design FMEA: Product functions; Materials and geometry; Design prevention.

Process FMEA: How production introduces defects; 6M causes; Feeds the Control Plan.

Prospective FMEA and retrospective RCA

First perspective: Asks: what could fail?; Acts before the next loss; Uses functions, experience and change knowledge.

Contrast or design check: Asks: what happened and why?; Acts after a failure signal; Verified findings should update the FMEA.

Source note: FMEA anticipates credible future failure; RCA explains evidence from an observed event. Verified RCA learning must update the FMEA.

Seven-step FMEA method

Plan & prepare → Structure → Function → Failure → Risk → Optimise → Document results

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Source note: This guide uses the seven-step planning, structure, function, failure, risk, optimisation and documentation sequence described in current AIAG/VDA-style practice.

Step 1 — planning and preparation

Purpose: Name the decision the FMEA must support.

Scope: Set product/process boundaries and interfaces.

Team: Include design, process, quality, maintenance and operators.

Inputs: Gather flow maps, complaints, controls, changes and lessons.

Steps 2–3 — structure and function

Structure: Break the process into logical steps and focus elements.

Interfaces: Show upstream, downstream and support interactions.

Function: Use verb–noun language with a measurable requirement.

Requirement: State the expected output, tolerance or customer need.

Step 4 — build the failure chain

Function → Failure mode → Effect → Cause → Current controls

Write useful failure statements

Failure mode: Loss, partial, intermittent, excessive or unintended function.

Effect: What the next operation, system or customer experiences.

Cause: A specific mechanism or process condition that creates the mode.

Control: What prevents the cause or detects cause/mode before escape.

Prevention and detection controls

First perspective: Remove or reduce the cause; Standard work, robust design, interlock; Influences occurrence.

Contrast or design check: Reveal cause or failure before escape; Sensor, test, inspection, alarm; Influences detection ranking.

Step 5 — S, O and D describe different risk dimensions

Severity: consequence → Occurrence: likelihood → Detection: ability to catch before impact

Severity rating discipline

Customer effect: Rate the most serious credible effect for the failure mode.

Consistent scale: Use one agreed 1–10 table and record the rationale.

Safety/regulatory: Escalate high-severity concerns regardless of a modest RPN.

Design change: Detection alone does not reduce the consequence severity.

Occurrence rating discipline

Cause-specific: Rate the likelihood of the stated causal mechanism.

Use evidence: Prefer capability, defect, maintenance or comparable-history data.

Prevention matters: Strong prevention can reduce occurrence.

Assumptions visible: Record uncertainty and the basis for the score.

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Detection rating discipline

Higher means weaker: A high D score means the failure is hard to detect.

Control-specific: Rate the actual detection control, not intended inspection.

Before effect: Ask whether the cause or mode is caught before customer impact.

Evidence: Use validation, gauge reliability and escape history.

RPN is useful—but incomplete

Calculate: RPN = S × O × D; Range 1–1000; Useful for sorting and trending.

Interpret: Ordinal ratings; Same RPN can hide different profiles; Record scoring rationale.

Guardrail: No fixed cutoff as permission; Severity first; Use team judgement and Action Priority.

Source note: Severity, occurrence and detection are ordinal judgements. Similar RPN values can represent very different risk profiles, so severity and action priority remain visible.

Same number, different risk

First perspective: S=10, O=2, D=2; RPN = 40; Potentially safety-critical effect.

Contrast or design check: S=5, O=4, D=2; RPN = 40; Different consequence and action logic.

Step 6 — optimise in the right order

Eliminate or reduce the effect → Prevent the cause / reduce occurrence → Improve detection → Assign owner and due date → Verify then re-rate

Strong recommended actions

Specific: Names the change, location and failure mechanism.

Owned: One accountable owner and a realistic due date.

Verifiable: Defines evidence that proves implementation and effectiveness.

Traceable: Links to the PFMEA row, change record and Control Plan.

Step 7 — document and sustain

Record decisions → Confirm actions → Re-score verified state → Update Control Plan → Set review triggers

Review triggers for a living FMEA

Engineering change: New material, equipment, tooling, software or parameter.

Process signal: Capability loss, special cause, defect trend or downtime.

Customer signal: Complaint, return, field failure or changed requirement.

Learning: RCA, audit, maintenance finding or supplier feedback.

FMEA pitfalls

Scope creep: A boundary too broad prevents meaningful depth.

Template completion: Scoring without insight or action creates false assurance.

RPN gaming: Adjusting ratings to cross a threshold hides risk.

Missing operators: The team loses actual-work failure knowledge.

No ownership: Actions without owner/date remain recommendations.

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Stale document: A frozen FMEA stops reflecting the real process.

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Activity 05 — Build the Orion Process FMEA

Mapping	A5, A6, K3
Objective	Apply structure, function and failure analysis to the scoped SMT process.
Deliverable	PFMEA header, process/function structure and at least eight complete failure chains
Tools/files	PFMEA Excel template · S/O/D rating guides · evidence references

Scenario

The team must turn the SIPOC, process flow, defect evidence and RCA findings into a traceable PFMEA.

Detailed Step-by-Step Procedure

Step 1 — Complete PFMEA purpose, scope, boundary, team, assumptions, revision and review trigger fields.

Work in: Orion-PFMEA.xlsx · Scope

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Import the five process steps and write verb–noun functions with measurable requirements.

Work in: Orion-PFMEA.xlsx · PFMEA

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — For each function, describe credible loss, partial, intermittent, excessive or unintended failure modes.

Work in: Orion-PFMEA.xlsx · PFMEA

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Write next-operation, system and customer effects; use the most serious credible effect for severity.

Work in: Orion-PFMEA.xlsx · PFMEA

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Write specific causal mechanisms using RCA evidence and the 6M view.

Work in: Orion-PFMEA.xlsx · PFMEA

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — Separate current prevention controls from current detection controls and cite evidence where available.

Work in: Orion-PFMEA.xlsx · PFMEA

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Scenario Questions

9. Does each failure mode describe loss of a stated function?
10. Which cause-control pair lacks evidence?

Acceptance Criterion

Every row forms a coherent function–mode–effect–cause chain, controls are correctly classified, and evidence references are traceable.

Submission Check

- □ All required tabs or sections are complete.
- □ Formula results reconcile to their source data.
- □ Evidence and assumptions are traceable.
- □ Both scenario questions are answered.
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Activity 06 — Rate Risk, Plan Actions and Re-score

Mapping	A5, A6, K3
Objective	Prioritise PFMEA actions using severity-first judgement, RPN context and verified re-rating.
Deliverable	Rated PFMEA with RPN formulas, internal action-priority guardrail, owners, due dates and post-action scores
Tools/files	Excel formulas · conditional formatting · action register · verification evidence

Scenario

Several Orion rows have similar RPNs but different customer and safety consequences.

Detailed Step-by-Step Procedure

Step 1 — Use the rating guides to assign S, O and D for each cause-control combination; record one-line rationales.

Work in: Orion-PFMEA.xlsx · Rating Guides

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Check that detection is rated in the correct direction: 10 means weak detection capability.

Work in: Orion-PFMEA.xlsx · PFMEA

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Review high-severity rows before sorting by calculated RPN; compare rows with equal RPN but different profiles.

Work in: Orion-PFMEA.xlsx · Risk Review

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Prioritise actions in order: eliminate/reduce effect, prevent cause/reduce occurrence, then improve detection.

Work in: Orion-PFMEA.xlsx · Actions

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Assign one owner, due date, implementation evidence and effectiveness measure to each action.

Work in: Orion-PFMEA.xlsx · Actions

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — After verification evidence is available, re-rate only the dimensions actually changed and compare residual risk.

Work in: Orion-PFMEA.xlsx · Re-score

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Scenario Questions

11. Why can an RPN of 40 require more urgent action than an RPN of 108?
12. Which rating should change after adding only a better inspection?

Acceptance Criterion

All RPN formulas are correct, severe risks remain visible, actions have ownership/evidence, and post-action ratings match the control change.

Submission Check

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Topic 4 — Identify and Direct Improvement Opportunities

Alignment: ELO4 · A7 · A8 · K2

Focus: Prioritise · select · pilot · coach · implement

Topic Orientation

Target the failure mechanism

Prevention is stronger than inspection when the cause can be removed or constrained.

Choose transparently

Decision criteria make trade-offs visible and reduce preference-driven selection.

Pilot before scale

A bounded test checks benefit, side effects, capability and adoption.

Direct with ownership

Every action needs an owner, due date, success measure and escalation path.

Concept Guide

The best action changes the failure mechanism

Inspection may catch a defect; prevention changes why it occurs.

From risk to improvement portfolio

High-priority risk → Countermeasure options → Decision criteria → Pilot and evidence → Approved implementation

Countermeasure hierarchy

Prevent: Eliminate the cause; Robust design; Poka-yoke or interlock.

Reduce: Improve capability; Control variation; Standard work and maintenance.

Detect: Automated checks; Validated inspection; Fast containment signal.

Opportunity sources

PFMEA: High severity, Action Priority, occurrence and weak detection.

Pareto and RCA: Recurring defects and verified mechanisms.

Capability and SPC: Unstable or incapable critical characteristics.

People and customers: Operator constraints, complaints and near misses.

DMAIC provides a project roadmap

Define: Clarify the customer, problem, scope, target and ownership.

Measure: Validate definitions and establish a trustworthy baseline.

Analyse: Prioritise patterns and verify causal mechanisms.

Improve: Select, pilot and implement countermeasures.

Control: Monitor, react, standardise and transfer ownership.

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Relationship to PDCA: Both are evidence-led cycles; DMAIC gives a detailed improvement-project structure.

5S supports stable process conditions

Sort: Remove items that are not needed for the work.

Set in order: Make the correct item and sequence obvious.

Shine: Clean while inspecting for early abnormality.

Standardise: Define a repeatable condition and visual method.

Sustain: Audit, coach and correct drift as normal work.

DOE tests factors and interactions

First perspective: Plan combinations of factor levels; Estimate main effects and interactions; Use randomisation, replication and blocking where appropriate.

Contrast or design check: Do not change many factors informally; Confirm measurement and process stability first; Use statistical expertise for safety-critical or complex designs.

Evaluation criteria

Risk reduction: Expected change to severity, occurrence or detection.

Effectiveness: Strength of evidence and fit to the causal mechanism.

Feasibility: Skills, time, technology and operational constraints.

Cost and speed: Investment, lifecycle cost and time to benefit.

Side effects: New failure modes, workload or compliance concerns.

Sustainability: Ability to standardise, train, monitor and maintain.

Weighted criteria matrix and Pugh matrix

First perspective: Scores options against weighted criteria; Useful when priorities differ in importance; Shows a total and sensitivity to weights.

Contrast or design check: Compares each option with a datum; Uses better / same / worse reasoning; Useful for concept screening and discussion.

Cost-benefit without false precision

Benefit: Avoided scrap, rework, downtime, complaints and risk exposure.

Cost: Capital, validation, training, maintenance and transition loss.

Time: Separate one-off savings from recurring annual impact.

Uncertainty: Show assumptions and ranges instead of one confident number.

Pilot learning cycle

Hypothesis and success measures → Safe scope and baseline → Run with controls → Compare result and side effects → Decide scale / revise / stop

A credible pilot

Representative: Covers the relevant product, shift and failure conditions.

Contained: Limits customer and production risk during the test.

Measured: Uses defined input, process and outcome measures.

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Reversible: Includes a stop rule and recovery plan.

Poka-yoke strategies

Contact: Physical fit or sensor; Detects shape/position; Prevents wrong assembly.

Fixed value: Counts required items or actions; Signals omissions; Useful for kits and fasteners.

Motion sequence: Enforces task order; Interlock or checklist; Prevents skipped steps.

Direct and coach the team

Purpose: Explain the customer/risk reason for the change.

Practice: Demonstrate, observe and let the operator perform.

Feedback: Correct the method and invite actual-work concerns.

Follow-up: Check adoption, outcome and unintended effects.

Implementation control

Owner and date: One accountable person and milestone.

Change control: Approved specification, work instruction and training.

Verification: Evidence the action was installed as intended.

Effectiveness: Evidence it reduced risk without creating new failure.

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Activity 07 — Select the Improvement Portfolio

Mapping	A7, A8, K2
Objective	Identify and compare risk-reduction opportunities using transparent criteria.
Deliverable	Opportunity register, weighted decision matrix, Pugh matrix and cost-benefit summary
Tools/files	Excel decision analysis · sensitivity check · PFMEA traceability

Scenario

Orion can fund only two of five proposed improvements during the next production quarter.

Detailed Step-by-Step Procedure

Step 1 — Translate PFMEA actions into five distinct options that address a named failure mechanism.

Work in: Improvement-Selection.xlsx · Options

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Agree weights for risk reduction, effectiveness, feasibility, speed, cost, side effects and sustainability.

Work in: Improvement-Selection.xlsx · Criteria

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Score every option with evidence notes; calculate weighted totals using formulas.

Work in: Improvement-Selection.xlsx · Weighted Matrix

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Choose the current method as the Pugh datum and compare each option as better, same or worse by criterion.

Work in: Improvement-Selection.xlsx · Pugh

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Estimate benefit and cost as ranges, including validation, training and maintenance—not only purchase cost.

Work in: Improvement-Selection.xlsx · Cost Benefit

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — Run a weight sensitivity check and recommend two actions with risks, dependencies and assumptions.

Work in: Improvement-Selection.xlsx · Recommendation

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Scenario Questions

13. Does the top score remain top when cost weight doubles?
14. Which option prevents the cause rather than merely detecting the defect?

Acceptance Criterion

The recommendation traces to PFMEA risks, calculations are formula-driven, assumptions are visible and trade-offs are explicitly discussed.

Submission Check

- □ All required tabs or sections are complete.
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Activity 08 — Pilot, Direct and Coach the Change

Mapping	A7, A8, K2
Objective	Plan a controlled pilot and direct team implementation with coaching and change safeguards.
Deliverable	Pilot charter, before/after measurement plan, risk guardrails, training and coaching checklist
Tools/files	PDCA pilot · change control · MOC checklist · coaching observation

Scenario

Orion will pilot an automated stencil-cleaning trigger and first-piece verification on Printer P2.

Detailed Step-by-Step Procedure

Step 1 — State the intervention and causal hypothesis: cleaning trigger reduces aperture blockage and bridge defects.

Work in: Pilot-Coaching.xlsx · Charter

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Define primary outcome, leading measures and guardrails; retain consistent operational definitions.

Work in: Pilot-Coaching.xlsx · Measures

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Select representative products, shifts and duration; define stop/containment rules and rollback.

Work in: Pilot-Coaching.xlsx · Design

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Update the temporary work instruction and identify training, access and approval requirements.

Work in: Pilot-Coaching.xlsx · Change Control

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Plan explain–demonstrate–practice–observe–feedback coaching for operators and shift leaders.

Work in: Pilot-Coaching.xlsx · Coaching

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Step 6 — Define the review decision: scale, revise or stop based on effectiveness, side effects and adoption.

Work in: Pilot-Coaching.xlsx · Decision

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Scenario Questions

15. What would cause the pilot to stop immediately?
16. How will the coach distinguish non-adoption from an ineffective countermeasure?

Acceptance Criterion

The pilot is measurable, contained and reversible; the change has approvals, training and a documented scale/revise/stop rule.

Submission Check

- □ All required tabs or sections are complete.
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Topic 5 — Manage Improvement with a Process Control Plan

Alignment: ELO5 · A9 · A10 · K4

Focus: Variation · capability · SPC · reaction · review

Topic Orientation

Control the critical few

Translate high-risk PFMEA causes and characteristics into practical controls.

Know the variation

Common-cause variation needs system improvement; special causes need investigation.

React by design

A Control Plan must state who acts, what is contained and how the process restarts.

Close the loop

Control results, incidents and changes feed back into the PFMEA and standard work.

Concept Guide

Control is a feedback system

A stable process detects meaningful change, contains risk and learns before the customer does.

Common-cause and special-cause variation

First perspective: Built into the current system; Many small sources; Improve the process as a system.

Contrast or design check: Specific, unusual signal; Assignable event or condition; Investigate and remove the cause.

Stability and capability answer different questions

Stability: Is the process behaviour predictable over time?

Capability: Can a stable process meet specification consistently?

Sequence: Check stability before interpreting capability indices.

Customer: Specifications come from requirements; control limits come from process data.

Cp and Cpk

Cp: Potential capability; Specification width ÷ 6σ; Assumes centring.

Cpk: Actual centring-aware capability; Minimum distance to limit ÷ 3σ; Cpk ≤ Cp.

Interpret: Use stable data; Confirm distribution and measurement; Report assumptions.

A control chart separates signal from noise

Plot observations in time order, calculate a centre line and statistically derived control limits, then investigate non-random signals. Control limits describe observed behaviour; specification limits describe requirements.

Control limits and specification limits

First perspective: Calculated from process behaviour; Support stability decisions; Not changed to make the chart look better.

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Contrast or design check: Defined by customer/design requirement; Support conformance decisions; Do not describe natural process variation.

Control-chart signals

Point beyond a limit: Immediate special-cause investigation.

Run on one side: Possible process shift.

Trend: Gradual change, wear or drift.

Cycle or clustering: Time pattern, batching or stratification.

The Control Plan operationalises the PFMEA

Process step / characteristic → Specification and method → Sample size and frequency → Owner and record → Reaction plan

Control selection principles

Risk-based: Strength and frequency reflect severity, occurrence and escape risk.

At source: Control as close as possible to the causal process condition.

Capable measurement: The method can reliably distinguish acceptable from unacceptable.

Actionable: A signal must trigger a defined response, not only a record.

Reaction plan

Stop or contain → Identify affected product → Notify owner → Correct and verify → Release and update records

Monitoring and evaluation

First perspective: Leading: parameter adherence, preventive maintenance, alarm rate; Detect drift before outcome loss; Useful for early intervention.

Contrast or design check: Lagging: defect, scrap, rework, return, downtime; Confirms realised outcome; Useful for accountability and learning.

Evaluate an improvement

Effectiveness: Did the targeted measure improve against baseline?

Sustainment: Did the result persist across time, shift and product mix?

Side effects: Did another defect, delay or workload appear?

Adoption: Is standard work followed and understood?

Risk: Are PFMEA ratings and controls updated with evidence?

Learning: What should be standardised, transferred or investigated next?

Closed-loop improvement

Monitor control data → Detect signal → Contain and investigate → Update PFMEA and action → Revise Control Plan and standard work

Governance review questions

Right risk?: Are high-severity and high-priority items visible?

Right evidence?: Are ratings, capability and results based on trusted data?

Right action?: Does the countermeasure address the verified mechanism?

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Right control?: Will drift be detected and contained in time?

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Activity 09 — Evaluate Stability and Capability

Mapping	A9, A10, K4
Objective	Monitor improvement performance using time order, control limits and capability measures.
Deliverable	Time-series dataset, control chart, signal log, capability calculation and evaluation statement
Tools/files	Excel AVERAGE · STDEV.S · UCL/LCL formulas · Cp/Cpk · trend interpretation

Scenario

After the pilot, Orion has 30 daily solder-paste-volume samples and bridge-defect results.

Detailed Step-by-Step Procedure

Step 1 — Review sample definition, measurement method, specification limits and time order.

Work in: SPC-Capability.xlsx · Data Dictionary

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — Calculate centre line and three-sigma control limits from the baseline period with formulas.

Work in: SPC-Capability.xlsx · Control Chart

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Plot observations in time order and flag points beyond limits, runs, trends or cycles.

Work in: SPC-Capability.xlsx · Control Chart

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Investigate each signal using shift, machine, material and maintenance context; record disposition.

Work in: SPC-Capability.xlsx · Signal Log

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — If the process is stable, calculate Cp and Cpk from specification limits, mean and standard deviation.

Work in: SPC-Capability.xlsx · Capability

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 6 — Write an evaluation covering effectiveness, sustainment, side effects, data limits and next action.

Work in: SPC-Capability.xlsx · Evaluation

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Scenario Questions

17. Why can a stable process still be incapable?
18. What does Cpk below Cp reveal?

Acceptance Criterion

Limits are formula-driven, signals are not confused with specification failures, capability is interpreted with stability and measurement caveats.

Submission Check

- □ All required tabs or sections are complete.
- □ Formula results reconcile to their source data.
- □ Evidence and assumptions are traceable.
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Activity 10 — Create the Process Control Plan and Review Loop

Mapping	A9, A10, K4
Objective	Translate PFMEA risks into operational controls, reaction plans and FMEA review triggers.
Deliverable	Process Control Plan, reaction plan, audit checklist, control-to-PFMEA trace matrix and review calendar
Tools/files	Excel Control Plan · PFMEA traceability · reaction plan · sustainment audit

Scenario

Orion must hand the improved SMT process back to production without losing the risk knowledge created by the team.

Detailed Step-by-Step Procedure

Step 1 — Select the critical process/product characteristics from high-priority PFMEA rows and the pilot evidence.

Work in: Orion-Control-Plan.xlsx · Traceability

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 2 — For each characteristic, state specification/target, method, sample size, frequency, owner and record.

Work in: Orion-Control-Plan.xlsx · Control Plan

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 3 — Write a reaction plan covering stop/contain, suspect-product boundary, notification, correction, verification and release.

Work in: Orion-Control-Plan.xlsx · Reaction Plan

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 4 — Check the measurement method, training and escalation are practical for all shifts.

Work in: Orion-Control-Plan.xlsx · Readiness

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Step 5 — Link each control row to PFMEA row ID and identify whether it prevents a cause or detects cause/mode.

Work in: Orion-Control-Plan.xlsx · Traceability

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

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Step 6 — Set monthly review plus trigger-based review for change, capability loss, incident, complaint or new RCA learning.

Work in: Orion-Control-Plan.xlsx · Review Calendar

Record the supporting evidence, assumption and decision rationale in the indicated file. If evidence is incomplete, label it as open and assign the next verification action rather than presenting it as confirmed.

Scenario Questions

19. Which PFMEA risk has no adequate operational control?
20. What event triggers immediate PFMEA review?

Acceptance Criterion

Every critical PFMEA risk has an owned control and reaction plan, traceability is complete, and review triggers close the learning loop.

Submission Check

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Optional tool: System Loop Builder

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Assessment Readiness

The Written Assessment contains four open-ended questions mapped to K1–K4 and takes 60 minutes. The Practical Performance contains five connected tasks mapped to A1–A10 and takes 90 minutes.

Explain the reasoning behind your decisions. A correct-looking template without a coherent failure chain, reliable formulas, evidence, ownership and control is incomplete.

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